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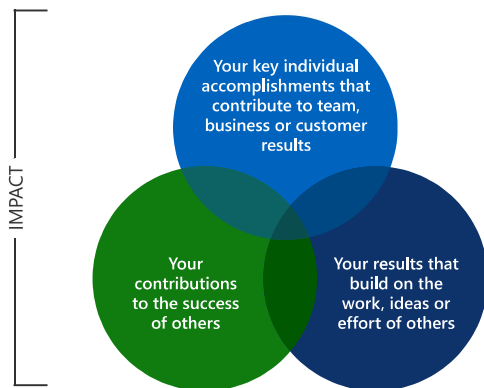
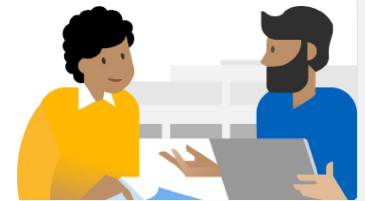
DEVESH KUMAR

SOFTWARE ENGINEER II | Reports to: Suma Kollipara

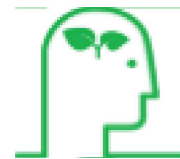
Connect Apr 2023

Reflection Period: Oct 1, 2022 - Apr 29, 2023

Connects provide a regular touchpoint where you and your manager can come together with a growth mindset to reflect on impact and discuss areas for improvement and future development.



Fixed Mindset leads to a desire to look smart and therefore a tendency to avoid challenges, give up easily, see failures as fruitless or worse, ignore useful negative feedback and feel threatened by the success of others.



Growth Mindset leads to a desire to learn and therefore a tendency to embrace challenges, persist in the face of setback, see failures as essential to mastery, learn from criticism and find lessons and inspiration in the success of others.

Reflect on the past

Capture what you've accomplished and learned since your last Connect.

Summarize your impact

Core priorities

What impact did you have for each of your core priorities?

Coding work:

- **Unit tests for Nov private preview :** Oct, 22
 - Initiated initial projects for I0 for Nov private preview.
 - Collaborated with Pragma to write tests for service as well as the controller projects. This would enhance product quality and help to mitigate bugs early.

- **Register Inbuilt entities job:** *Jan, 23*
 - Wrote startup jobs for registering inbuilt entities. These startup jobs created MDS healthdoc which would indicate users whether the startup jobs were successful or not.
 - Wrote code for registering register entity to azure digital twin by fetching their DTDL models through the URL in the payload. These were the default entities which would be used by user to ingest data.
 - Wrote a tool (**register entities payload tool**) for creating payloads for inbuilt entities. This helped to greatly reduce developer time to create payloads for inbuilt entities as there were about 80 entities. This tool also enabled program manager team to restructure the payload and entities frequently. This tool removed the aspect of human error that would occur while writing payload for about 80 odd entities.
- **Create, delete and Get API** *Jan, 23*
 - Added API for creating and deleting register entities. This API would enable users to register custom entities and to further use its schema to ingest data.
 - Added get API for fetching the health doc to see the status of the MDS startup job and get API to fetch individual registered entity.
 - Wrote unit test for all the function. Also wrote L1 test for the controller.
- **Making ingestion project work in Azure Function** *Feb23*
 - Made changes in the ingestion work in Azure functions. Made accurate bicep changes so that it had all the permissions to access all the resources.
 - Migrated the dependency of the project from app setting to function app config. Wrote all the config setting which were present in appsetting to the function app config. Also made changes in the code base to read from the function app config instead of appsettings.json and made equivalent bicep changes for the same.
- **Bicep** *Feb23*
 - Added and modified existing bicep files to incorporate new resource creation.
 - Streamed lined different storage account used across the solution to three storage account one for background jobs, one for azure functions and one for batch ingestion.
- **Virtual network and private endpoints** *Feb23*
 - Moved all the resource containing user data behind virtual network. This greatly improved the security for our solution as no outside connections can be made.
 - Made private dnsZones and private endpoints for the all the resource to talk to each other. Also fixed the outbound connections of the resources to go through the virtual network.
 - Divided the virtual network into subnet for different types of resources like funcsubnet etc.
 - Made managed private endpoints to access the resources from azure data explorer.
 - At last made sure that everything worked in same way before and after the virtual network implementation.

Technical debt Item and codebase improvements

- **Remove warnings** *Mar, 23*
 - Removed about **1152 warnings** from the codebase. These were pointed the MIS team as tech debt item.
 - The warnings were scattered throughout the codebase from the unit test to main projects and were comprised of null exceptions and unnecessary use of async keyword.
 - Made changes in the config folder of projects to treat them as error from next time. This will make sure that we do not have warnings in future.
 - This greatly improved the code quality of our codebase. And this will decrease chance of any null error in future.
- **Standardize Exception handling** *April, 23*
 - Standardize Exception handling according to https standards. This streamlines different exceptions to a common format and gives all the relevant information.
- **L0 coverage check** *April, 23*
 - Added code coverage check in the azure pipeline for our repository. This will automatically fail the pull request if the l0 coverage is below 70 percent. This will help ensure that every line of code pushed is of great quality.

POC work:

- **Function App** *Nov, 23*
 - Did spike to get familiarized with the working of azure function. Explored different triggers used in function app.
 - Used cosmos trigger of the function app to read the data from the cosmos and write onto the event hub. This was a part of a larger demo where we as a team showed how through function app we can write data in Event-hub and further use this data in generate graphs.

- Configured the demo projects to use the managed identities and in the function app. Made appropriate bicep changes for the same.
- **Azure data explorer and Continuous export** *Dec. 23*
 - Collaborated with Shubham to explore how continuous export feature of the azure data explorer can be used to transfer cold data. Through this feature we could save customer money as storing in adx is expensive.
 - Wrote a one pager of my finding. This feature was later implemented in our solution.
- **Alternate approach to mimic hops** *April, 23*
 - Did POC to mimic hops in azure data provider. Earlier we wrote query on azure digital twin to get the hops. As the we had the similar data stored in azure data explorer, i explored how can we mimic this in azure data explorer.
 - Tested the performance of query on both data stored in ADX and the cold data (stored in ADLS).
 - The results in azure data explorer were significantly faster (about 5 -6 times faster). Wrote one pager about my finding [here](#).

Bugs and fixes:

- **Function app Relationship flow**
 - Found and fixed a bug that was causing relationship flow of the ingestion function to fail.
- **Consumption flow**
 - Found and fixed a bug that was giving status 401 unauthorized in some consumption query.
 - There was a missing permission required by azure data explorer to azure digital twin. Corrected bicep template to resolve it
- **Cosmos key fetching issue after enabling locks**
 - Found a bug in which cosmos key was coming null if locks were implemented and we hit the API.
 - Resolved the bug by fetching the keys before the locks were implemented.

Diversity & Inclusion (D&I) Core Priority

What impact did your actions have in contributing to a more diverse and inclusive Microsoft?

- Our team is scattered around different places and time zones. There are people who come to office and some people who work from home for some days. When asking for help or any spec requirements I try to incorporate their timezones and do not ping them outside their work hours. I often used the "send at scheduled time" feature of teams to send the messages and replies at their relevant time zones.



Manager comments

Great job on all the work you have done! Your efforts have significantly improved the quality and efficiency of the product.

Your work on unit tests for the Nov private preview and collaborating with Pragya to write tests for service as well as the controller projects is commendable. This will **undoubtedly bolster the product's quality and reduce bugs early on**.

The tool you developed for creating payloads for inbuilt entities is impressive. It has greatly **reduced developer time and enabled the program manager team to modify payloads and entities frequently while eliminating human errors**.

The virtual network and private endpoints implementation, dividing the virtual network into subnets, and making managed private endpoints to access the resources from azure data explorer were great enhancements to **improve the solution's security**.

The removal of warnings from the codebase and standardizing exception handling according to https standards have greatly improved the code quality. Adding code coverage check in the azure pipeline for our repository will help ensure that every line of code pushed is of great quality.

Your POC work on Function App, Azure data explorer and Continuous export, and Alternate approach to mimic hops has added value to the solution, and it's great that the continuous export feature of the azure data explorer was implemented.

Your contribution to fixing bugs in the Function app Relationship flow, Consumption flow, and cosmos key fetching issue after enabling locks is impressive.

Overall, your contributions are commendable, and your work has significantly improved the product quality and efficiency. Keep up the excellent work!

To have a more significant impact in the upcoming semester, I encourage you to contribute to the success of others by mentoring interns, unblocking any obstacles they face, and supporting them in achieving their goals.

Reflect on a challenge or setback

Growth mindset is about learning from our mistakes and setbacks to enable our future growth. It's important that we learn and adapt quickly. Then, we can apply what we've learned so we can continually grow and deliver greater impact for our teams, customers, and business.

Consider when you could have done something differently. How will you apply what you learned to make an even greater impact?

I would have done these things differently:

- While removing the warnings I did not make treat warnings as error flag true. This resulted in resurfacing of some warnings and lead to some double work.
- For the bicep work it was a lot of hit and trial work. If better documentation for the bicep templates and virtual network available things could have done quickly.
- For the register entity flow there was a lot of changes in the payload for the inbuilt entity. Things here could have moved quickly if we could have fixed the flow and the payload before.



Manager comments

It's great that you have identified areas where you could have done things differently and that you are open to improving your work.

Plan for the future

Think about opportunities for impact in the upcoming Connect period.

Review your core priorities

Core priorities

What are your 3-5 core priorities?

1. Enhancing codebase by focusing on technical debt items

Focus areas for the upcoming period to drive business impact:

1. Few code enhancements were pointed out by our sister team. I would work on implementing them to improve security and coding practice.
2. To resolve any bug that pop up after the private preview to enhance customer experience.
3. Participate in more pull request reviews raised by other developer which would enable me to have better understanding of overall solution.
4. Adding more unit test cases and improving test coverage.

Success measures:

- Less bugs faced by customer and better maintainability of codebase.
- Improved security of the solution
- Feedback from customer and customer adoption rate would help in determining effectiveness of the new features
- For tests a better code coverage result.

2. Enhancing Consumption APIs

Focus areas for the upcoming period to drive business impact:

1. Adding more feature in consumption project like data historization filters. Though data historization customer will be able to query data through time.
2. Using more ADX to query data instead of ADT and cosmos as it is faster.
3. Adding multi hop feature in relationship API. This will enable user to query relations that are more than 2 edge away from the source node.
4. Analyze and review new resources that could be used in our solution that would provide customer cheaper and better experience

Success measures:

- Reduced latency of the existing APIs.
- With multi hop a customer would be able to do root cause analysis more efficiently and faster.

Diversity & Inclusion (D&I) Core Priority

To realize our mission to empower every person and organization on the planet to achieve more, each and every one of us must help create a diverse and inclusive culture where everyone can bring their full and authentic self, where all voices are heard, and where we do our best work as a result. With this priority, personally commit to be an active and intentional participant in creating a diverse and inclusive Microsoft.

Select and commit to at least one focus area from the options below, or create one of your own.

- Seek diverse perspectives and incorporate feedback, apply insights from a variety of sources and people
- Mentor and/or support a person who may benefit

How will you define success as you activate your D&I Core Priority?

- The perspective and feedback from others would be a good measure for success.



Manager comments

A good set of priorities to follow up on in the upcoming period. One of the key objectives for the SKG team is to improve the consumption APIs, which significantly affects the AI copilot experience. Additionally, it is recommended to regularly seek feedback from the product team to monitor progress and make course corrections if necessary.

Identify your opportunities for growth

Consider what more you want to learn and how you want to grow in your current role.

What are the 1-2 actions you will take to energize your learning, growth, and development over the next period?

- I will try to volunteer for some activities apart from the traditional work like team building activities
- Ask for more feedback and code reviews of my pull requests.
- Learn more about manufacturing and built a better domain knowledge.
- Focus more on additional learning some new technologies particularly large language model.

What support do you need to fulfill these actions? Fill in any that apply.

Opportunities to (optional):

No Response

Perspective or coaching on (optional):

On my coding practices and feedback/ improvements in my pull requests.

Learning on (optional):

To learn more about openAI work and get a hands on experience on it



Manager comments

In alignment and looking good.

Conversation starters (optional)

Choose any of these optional prompts to continue great conversations with your manager. Write a brief response (1-15 words).

Of my strengths, the ones I'd like to apply more in my work are:

In my work, I feel motivated or energized by:

In my work, I lose motivation or energy from:

I know I can't do everything on my own. I'd like help with:

It can be challenging to prioritize important topics for 1:1 conversations. Plan ahead and flag any of the following topics you'd like to discuss with your manager in upcoming conversations.

Select any conversation topics below, or add your own:

- Career:** Discuss ways to grow my career to align with my strengths, interest & aspirations.
- Learning:** Explore potential development and growth opportunities to optimize your performance.
- Feedback:** Get input on how I'm doing in my role and how to deliver great impact.

Check out aka.ms/career for resources and guides on developing your career.



Manager comments

I have observed that devesh is most energized when working on challenging and complex tasks that require critical thinking and problem-solving skills. He also enjoys opportunities to learn new skills and technologies.